

Why the Year Is Cut into Twenty-Four Parts

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Abstract

The year is cut into twenty-four solar terms, and the cut is not arbitrary. The national standard GB/T 33661-2017 defines a solar term as an instant at which the apparent geocentric ecliptic longitude of the Sun equals an integral multiple of fifteen degrees; three hundred and sixty divided by fifteen gives twenty-four. Zero degrees of the ecliptic is fixed at the vernal equinox, so the terms run from 0° at the equinox to 270° at the winter solstice and back. Because the Gregorian calendar and the solar terms measure the same body, the Sun, their graduations line up and the Gregorian dates of the terms drift by only a day or two from year to year, while their dates in the lunar calendar swing by as much as a month. The twenty-four names fall into two batches: eight that mark the Sun's key stations, and sixteen that describe finer changes of climate. They divide again into two groups of twelve, the jie at the head of the lunar month and the qi at its middle; in four-pillar reckoning only the jie bounds the month pillar. A further distinction separates qi, the rhythm of the seasonal qi used on the chart, from hou, the visible state of plants and weather, which is established about a month behind qi. The word qi itself carries at least three senses, and the note separates them. The earliest document to record the complete list of twenty-four names is the *Huainanzi*, chapter "Tianwen xun", of the Western Han; the *Shiji* refers to the "twenty-four sections" without listing them. The note is an exposition of calendrical facts only. It contains no predictive claims and no individual prognostication.

Keywords: twenty-four solar terms; solar longitude; ecliptic; GB/T 33661-2017; jie and qi; hou; *Huainanzi*

Foreign-venue shell. The abstract, keywords and reference list are added for posting outside China. The Chinese original has none of them. Everything from § 1 onward is the original text, translated.

1. A Step of Fifteen Degrees

The start of spring never leaves the band between 3 and 5 February in the Gregorian calendar. Look through thirty years of almanacs and it changes place within those three days.

In the lunar calendar, though, the same start of spring sometimes falls in the twelfth month and sometimes in the first, a whole month apart.

One scale, fixed as a nail in one almanac and swaying to and fro in the other. That fact is itself the origin of the twenty-four solar terms.

The national standard GB/T 33661-2017, *Calculation and Promulgation of the Chinese Calendar*, defines a solar term bluntly:

Solar terms: the collective name for the twenty-four instants in a tropical year at which the apparent geocentric ecliptic longitude of the Sun equals an integral multiple of 15 degrees.

The key word is "instants".

A full circuit of the Sun along the ecliptic is 360 degrees. Cut into 24 parts, each is exactly 15 degrees. When the Sun reaches one of those fifteen-degree marks, that instant is a solar term. Three hundred and sixty divided by fifteen gives twenty-four; that is where the twenty-four parts come from.

Zero degrees of the ecliptic is fixed at the vernal equinox. Counting from the equinox, the equinox is 0 degrees; each term thereafter adds fifteen, until the winter solstice stands at 270 degrees, and a full circuit returns to the equinox.

One thing should be made clear. In astronomy the count starts from the vernal equinox; most popular tables of solar terms begin at the start of spring. The two reckonings coexist. The starting point differs; the names and their order are exactly the same. Before reading a table, see which end it starts from, and the count will not go wrong.

2. Table of the Twenty-Four Solar Terms

Term	Longitude	Gregorian (approx.)	English name
立春	315°	3–5 Feb	Start of Spring
雨水	330°	18–20 Feb	Rain Water
惊蛰	345°	5–7 Mar	Awakening of Insects
春分	0° (360°)	20–22 Mar	Spring Equinox
清明	15°	4–6 Apr	Pure Brightness
谷雨	30°	19–21 Apr	Grain Rain
立夏	45°	5–7 May	Start of Summer
小满	60°	20–22 May	Grain Buds
芒种	75°	5–7 Jun	Grain in Ear
夏至	90°	21–22 Jun	Summer Solstice
小暑	105°	6–8 Jul	Minor Heat
大暑	120°	22–24 Jul	Major Heat
立秋	135°	7–9 Aug	Start of Autumn
处暑	150°	22–24 Aug	End of Heat
白露	165°	7–9 Sep	White Dew
秋分	180°	22–24 Sep	Autumn Equinox
寒露	195°	8–9 Oct	Cold Dew
霜降	210°	23–24 Oct	Frost's Descent
立冬	225°	7–8 Nov	Start of Winter
小雪	240°	22–23 Nov	Minor Snow
大雪	255°	6–8 Dec	Major Snow
冬至	270°	21–23 Dec	Winter Solstice
小寒	285°	5–7 Jan	Minor Cold
大寒	300°	20–21 Jan	Major Cold

Longitudes are given as positive values, so the start of spring is written 315° rather than reached from the negative side. In the table jie and qi alternate: odd positions are jie, even positions are qi. That layer is taken up below.

The English names follow the internationally current set, which differs slightly from the set published by the China Meteorological Administration: the administration writes the start of

spring as "the Beginning of Spring" and grain buds as "Lesser Fullness of Grain". Both sets are in use; finding either is no error.

3. Why It Barely Shifts in the Gregorian Calendar

The Gregorian calendar and the solar terms measure the same thing: the Sun.

The Gregorian calendar is laid out by the tropical year; the solar terms are fixed by the Sun's position on the ecliptic. Two rulers set against the same reference, their graduations naturally line up. In the first half of the year the jie mostly fall near the 6th and the qi near the 21st; in the second half the jie near the 8th and the qi near the 23rd. The drift from year to year is no more than a day or two.

The lunar calendar is another matter. Its months are fixed by the synodic month, a little over twenty-nine and a half days; twelve such months fall eleven days short of a tropical year, and a leap month is added every few years. The dates of the solar terms in the lunar calendar therefore swing widely; the "double spring" and "springless" years arise in this way. This was covered earlier; it is mentioned here only in passing.

4. Two Batches Inside the Names

The twenty-four names were not all written down at once. Seen as two batches, they become clear.

The first batch is eight: 立春 (start of spring), 春分 (spring equinox), 立夏 (start of summer), 夏至 (summer solstice), 立秋 (start of autumn), 秋分 (autumn equinox), 立冬 (start of winter), 冬至 (winter solstice). These eight divide the Sun's key stations: the two 立 open the four seasons, the two 分 are the equal nights, the two 至 the longest and the shortest day.

The second batch is sixteen: 雨水, 惊蛰, 清明, 谷雨, 小满, 芒种, 小暑, 大暑, 处暑, 白露, 寒露, 霜降, 小雪, 大雪, 小寒, 大寒. These divide the finer changes of climate.

The wording itself shows where the two batches come from. Heat has a pair, 小暑 and 大暑; snow has a pair, 小雪 and 大雪; cold has a pair, 小寒 and 大寒; dew appears twice, as 白露 and 寒露. Both 雨水 and 谷雨 carry the word for water. Every one of these characters describes a scene on the ground, not a degree in the sky.

5. The Jie at the Head of the Month, the Qi at Its Middle

The twenty-four names divide once more, into two groups of twelve.

The twelve jie: 立春, 惊蛰, 清明, 立夏, 芒种, 小暑, 立秋, 白露, 寒露, 立冬, 大雪, 小寒.

The twelve qi: 雨水, 春分, 谷雨, 小满, 夏至, 大暑, 处暑, 秋分, 霜降, 小雪, 冬至, 大寒.

One jie with one qi makes a single lunar month. The jie stands at the head of the month, the qi at its middle.

In charting, the boundary of the month pillar recognises the jie and not the qi. The instant 立春 arrives, the month is 寅; the instant 惊蛰 arrives, it is 卯 — with no regard for whether the lunar date is the first of the month. This was explained earlier; the two sets of names are merely completed here. The twelve qi have other uses in charting.

节·月首

月柱的界线落在这十二个上

立春	惊蛰	清明	立夏	芒种	小暑
立秋	白露	寒露	立冬	大雪	小寒

气·月中

另有用途，不作月柱分界

雨水	春分	谷雨	小满	夏至	大暑
处暑	秋分	霜降	小雪	冬至	大寒

一个节配一个气，合成一个农历月

6. Qi and Hou: Two Rulers Half a Beat Apart

There is one layer of calendrical writing that is easily misread, and it has to be set out first.

Qi measures the rhythm of the seasonal qi: the wood qi enters in the 寅 month, is at its strongest in 卯, and yields in 辰. This ruler serves charting; the moment a solar term arrives the qi changes shift, to the second.

Hou measures what the eye sees: by the first month the month is already 寅, yet the cold is still heavy and the plants do not stir; only in 卯 do the grasses grow and the birds fly; only in 辰 is there real abundance. This ruler describes the visible face of the ground.

The two rulers are half a beat apart: hou is established about a month behind qi.

So the wood qi has already declined by the 辰 month, and you look up at the trees in the field and find 辰 only now at its fullest; the two statements do not conflict: one speaks of the qi on the chart, the other of the hou on the ground. The four characters 辰, 戌, 丑 and 未 are peculiar precisely because they stand at the point where the qi has turned and the hou has not yet caught up.

7. Three Uses of the Word Qi

The same character carries at least three meanings in this field; read it by context.

First, the middle qi (中气): the twelve names set against the jie, the group of 雨水, 春分 and the rest. This is the usage in charting.

Second, the length of one solar term, fifteen days. To say "one qi" is to mean that stretch of days.

Third, a method of dividing the year: 平气 and 定气. Division by equal time is 平气; division by the Sun's actual position is 定气. The third was treated in a piece of its own earlier and is not repeated here.

8. How the Ancients Measured Out the Twenty-Four Parts

The ancients had no notion of ecliptic longitude. They used the gnomon: a pole set upright, and the noon shadow measured.

At the winter solstice the shadow is longest, at the summer solstice shortest; the two poles are fixed first. With the two poles in hand, the interval is subdivided into twenty-four parts, and there is a usable scale of time.

The book that first wrote the twenty-four names out in full is the Huainanzi, chapter "Tianwen xun" (天文训), of the Western Han. Its sequence begins at the winter solstice:

斗指子，则冬至。

When the Dipper points to 子, it is the winter solstice. The twenty-four names are exactly the same as today's; this is the earliest complete list in the surviving literature. The Shiji, in the "Tai Shi Gong zi xu" (太史公自序), also passes over the "twenty-four sections" in a single sentence: yin and yang, the four seasons, the eight positions, the twelve measures and the twenty-four sections each have their ordinances; those who follow them prosper.

What the ancients measured was the length of a shadow; what we measure is the Sun's degree on the ecliptic. The two methods aim at the same set of instants; the later one merely grinds the

ruler finer.

As for how people lived before calendars appeared, that has a piece of its own; one sentence will serve here.

Closing

Once the twenty-four parts are fixed, the year has twenty-four rulers, each of fifteen degrees.

The most important use of these twenty-four rulers is to cut the year into twelve shares, one to a month. Whether the month pillar changes depends on whether the jie has arrived, and not on which page the calendar has reached. So check the moment of the jie before charting; get that wrong and everything after it is wrong.

What comes next in this series is another scale in the sky. The Sun's position fixes years and months; the Moon's fixes the new and full moons. The two scales together make what we today call the lunar calendar.

A&Q

Why are there exactly twenty-four solar terms?

The Sun travels 360 degrees along the ecliptic in a year, and a term falls every 15 degrees; 360 divided by 15 gives 24.

Why are the dates of the terms nearly fixed in the Gregorian calendar but widely swinging in the lunar calendar?

The Gregorian calendar and the solar terms both take the Sun as their reference, so the two rulers line up and the Gregorian dates drift by no more than a day or two. Lunar months are fixed by the synodic month of a little over twenty-nine and a half days, eleven days short of a tropical year, and the gap is closed by leap months; the dates of the terms in the lunar calendar therefore move widely back and forth.

What is the difference between a jie and a qi?

The twenty-four names form two groups: the jie at the head of the lunar month and the qi at its middle, one of each making a single lunar month. In charting, the boundary of the month pillar recognises the jie and not the qi.

Are qi and hou the same thing?

No. Qi is the rhythm of the seasonal qi, used in charting, and it changes shift the instant a solar term arrives; hou is the visible state of plants and weather, and it is established about a month behind qi.

In which book do the names of the twenty-four solar terms first appear complete?

The Huainanzi, chapter "Tianwen xun", of the Western Han. Its sequence begins at the winter solstice, and its twenty-four names are exactly the same as today's; it is the earliest complete list in the surviving literature.

Note by Chujia Kuang

The two numbers worth remembering from this piece are 15 and 24. Whenever a solar term comes up, ask first which fifteen-degree mark it falls on, then ask whether the day is a jie or a qi. Settle those two questions and the whole line is clear.

About the author

Chujia Kuang (邝楚嘉), pen name Jiayan Yide (嘉言一得), is an independent researcher in Jieyang, Guangdong, devoted to archiving verifiable primary sources and calendar data.

This column writes about calendrics, solar terms, the stems and branches and orientation, and also about what he reads. The pen name 嘉言一得 comes from the 史记: 愚者千虑，必有一得, "the foolish man's thousand thoughts must yield one gain". Where something is written wrongly, corrections are welcome.

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